Discrimination of Underground Nuclear Explosions from Natural Earthquakes Using Seismic Waveforms, Source Physics, and Regularized Machine Learning

Advanced Statistical Methods II Course Project Report

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Abstract

Distinguishing clandestine underground nuclear explosions from natural earthquakes is the central technical challenge in global non-proliferation monitoring under the Comprehensive Nuclear-Test-Ban Treaty (CTBT). In this project for Advanced Statistical Methods II, we construct an end-to-end statistical and deep learning pipeline utilizing full three-component (Z, N, E) seismic waveforms. We synthesize source physics—specifically compressional P-wave dominance, shear (S/L_g) wave absence, spectral decay, and 3D particle motion rectilinearity—into 18 discriminative features. We evaluate five regularized and ensemble statistical classifiers (Lasso L_1, Elastic Net, Random Forest, SVM with RBF kernel, and XGBoost) alongside a 3-Channel 2D Convolutional Neural Network (128×128 zero-padded spectrograms). Crucially, all models are tuned and validated using 5-Fold Group Cross-Validation grouped strictly by seismic event ID to eliminate spatial data leakage. Furthermore, we address the critical real-world problem of extreme class imbalance ($N_{\text{earthquakes}} \gg N_{\text{explosions}}$), mathematically demonstrating why ROC-AUC provides a dangerously optimistic assessment and showing why Precision-Recall AUC (PR-AUC) must be adopted for both model selection and evaluation. Our tuned models achieve outstanding out-of-event performance: SVM (RBF) and XGBoost achieve 97.13% and 96.55% accuracy respectively, while the 3-Channel CNN attains 97.98% accuracy, 99.01% precision, and 0.9976 PR-AUC. Finally, multi-station network consensus voting achieves 100% detection across all six historical North Korean underground nuclear tests held completely outside model training and tuning.

1 Introduction and Problem Formulation

Underground nuclear detonations generate seismic waves that propagate across continental and teleseismic distances, sharing basic observational similarities with natural tectonic earthquakes. Since the inception of the Comprehensive Nuclear-Test-Ban Treaty (CTBT), the International Monitoring System (IMS) has operated a global network of seismological stations to verify compliance.

From an applied statistical modeling perspective, seismic event discrimination presents three distinct challenges:

1. **High-Dimensional Time-Series & Non-Stationarity:** Raw seismograms are non-stationary, oscillatory time-series characterized by distinct wave phase packets (P_n, P_g, S_n, L_g, and surface waves) arriving across time.

2. **Spatial Correlation and Grouping Structure:** A single seismic event is typically recorded by multiple stations globally. A naive random split of station traces produces severe *data leakage*, artificially inflating validation accuracy because the model simply memorizes the event signature rather than learning generalized physics.

3. **Severe Class Imbalance in Operational Deployments:** In nature, tens of thousands of earthquakes occur annually, compared to zero or one clandestine nuclear tests. Under such severe imbalance, standard evaluation metrics like Accuracy and ROC-AUC become deceptive, as even a minuscule False Positive Rate generates an overwhelming volume of false alarms.

In this project, we address these challenges through a unified three-component (Z, N, E) seismic framework. We examine whether physics-guided feature extraction combined with regularized generalized linear models (Lasso, Elastic Net), modern tree/kernel ensembles (XGBoost, SVM), and end-to-end representation learning (3-Channel 2D CNN) can reliably separate these phenomena.

2 Historical Motivation

The verification of international peace and the enforcement of non-proliferation frameworks heavily rely on our ability to accurately monitor global seismic activity. The very birth of modern digital signal processing was forged out of the geopolitical necessity to differentiate between natural tectonic earthquakes and man-made underground nuclear explosions. During the Geneva nuclear test ban negotiations in the late 1950s and early 1960s, a major gridlock emerged: the Soviet Union refused on-site verification inspections, and the United States lacked a reliable way to distinguish low-yield underground weapon detonations from routine earthquakes using remote seismometers.

The solution lay in the frequency domain. The isotropic, concussive pressure of an explosion produces fundamentally higher frequency components and faster decay rates than the slow, shear friction of a shifting tectonic fault line. However, calculating the Discrete Fourier Transform (DFT) for thousands of continuous seismic samples required $\mathcal{O}(N^2)$ computational operations. The computational burden was so massive that contemporary computers lacked the processing power to filter and analyze seismic streams in real time. This bottleneck directly motivated the development of the Fast Fourier Transform (FFT) by James Cooley and John Tukey in 1965 [1]. By reducing the computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$, the FFT fundamentally unlocked the capability to digitally process and analyze global seismic frequencies rapidly enough to track nuclear compliance.

While the classic FFT processes an entire signal to give a static snapshot of its frequencies, modern verification tasks demand an understanding of how those frequencies change over time. This project builds directly upon the legacy of the Cooley-Tukey FFT by implementing the Short-Time Fourier Transform (STFT). By sliding a localized temporal window across continuous, three-component (3C) seismic data and computing the FFT at each step, we generate 2D time-frequency images known as spectrograms. This project scales these foundational mathematical breakthroughs from traditional algorithmic filtering into the domain of computer vision, exploring how modern 2D Convolutional Neural Networks (CNNs) can automatically interpret these STFT spectrograms to automate high-stakes nuclear monitoring.

3 Physical Foundations and Exploratory Waveform Analysis

Distinguishing underground nuclear explosions from natural earthquakes relies on fundamental differences in their source mechanics and the resulting seismic wavefield.

3.1 Source Mechanisms: Isotropic Expansion vs. Shear Slip

- **Underground Nuclear Explosion:** Modeled as an isotropic pressure pulse radiating outward from an explosive cavity. In the equivalent point-source approximation, the seismic moment tensor is purely diagonal (hydrostatic):

$$\mathbf{M}_{\text{explosion}} = M_0 \mathbf{I}_{3\times 3} = \begin{pmatrix} M_0 & 0 & 0 \\ 0 & M_0 & 0 \\ 0 & 0 & M_0 \end{pmatrix}$$

Because spherical expansion produces purely radial volume change ($\nabla \times \mathbf{u} \approx 0$), an explosion radiates primary compressional waves (P-waves) with virtually no direct shear wave (S-wave) excitation at the source.

- **Natural Earthquake:** Initiated by frictional shear slip across a planar fault with unit normal $\hat{\mathbf{n}}$ and slip vector $\hat{\mathbf{d}}$ ($\hat{\mathbf{n}} \cdot \hat{\mathbf{d}} = 0$). The resulting double-couple moment tensor is symmetric and traceless:

$$\mathbf{M}_{\text{earthquake}} = M_0(\hat{\mathbf{n}}\hat{\mathbf{d}}^T + \hat{\mathbf{d}}\hat{\mathbf{n}}^T), \qquad \text{tr}(\mathbf{M}_{\text{earthquake}}) = \sum_{i=1}^{3} M_{ii} = 0$$

Because shear dislocation involves no net volume change, it directly generates intense transverse shear waves (S, S_n, L_g) that oscillate perpendicular to the ray path, depositing substantial energy onto horizontal components (N, E).

3.2 Exploratory Analysis: Waveform Structure Comparison

To empirically investigate these physical differences, Figure 1 compares synchronized 3-component recordings (Z, N, E) of a representative US underground nuclear test (Nevada Test Site, Station MNV) against a moderate natural earthquake (EarthScope GSN, Station TUC). Both events are sampled at 20 Hz, bandpass filtered to $0.8 - 5.0$ Hz to isolate body waves from long-period microseisms, and normalized by each event's 3-component peak amplitude ($A_{\max} = \max(|Z|, |N|, |E|)$).

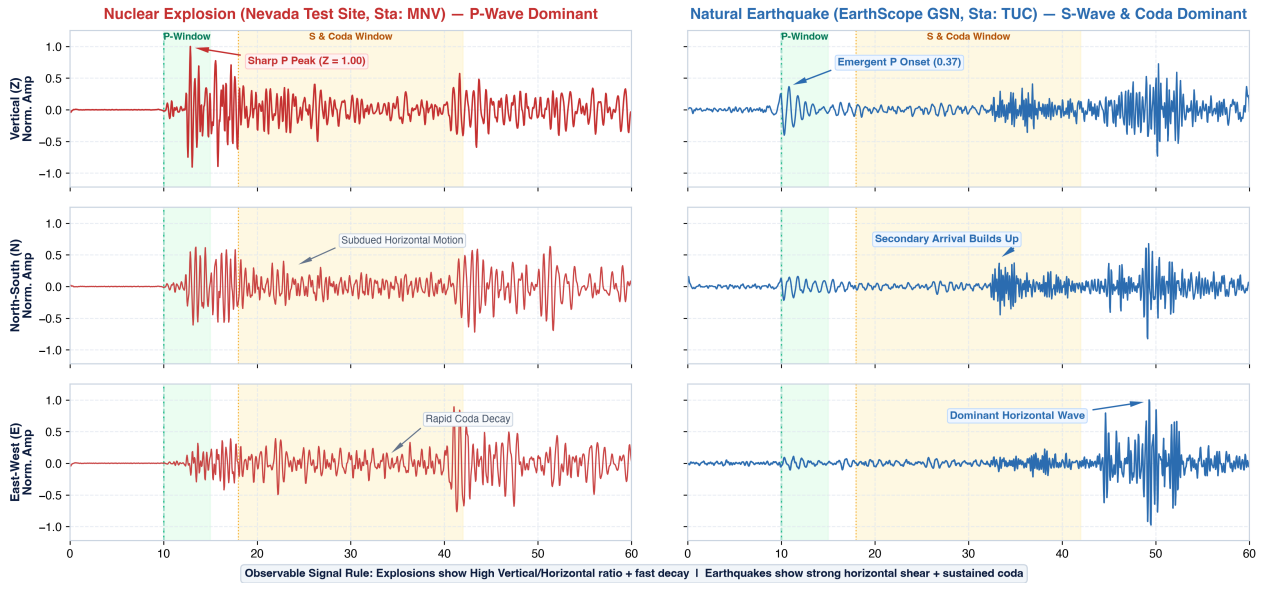


Figure 1: Exploratory Waveform Comparison between Nuclear Detonations and Natural Earthquakes. Synchronized 3-component recordings (Z, N, E) over a 60-second window. The left column shows a Nevada Test Site underground nuclear explosion (Station MNV); the right column shows a natural tectonic earthquake (Station TUC). Green shaded regions indicate the compressional P-wave window ($t \in [10.0, 15.0]$ s), with P-onset aligned at $t_P = 10.0$ s; amber shaded regions indicate the shear (S) and coda window ($t \in [18.0, 42.0]$ s). Explosions display an impulsive vertical P-wave with a severe horizontal shear deficit and rapid decay, whereas earthquakes exhibit an emergent P-wave followed by dominant horizontal shear shaking and prolonged coda.

The exploratory comparison reveals three pronounced structural contrasts:

1. **Compressional Dominance vs. Shear Wave Deficit:** In the explosion recording, wave energy is concentrated almost entirely on the vertical component (Z) as an impulsive P-wave arrival ($A_{\text{norm}} = 1.00$). The horizontal channels (N, E) exhibit minimal amplitude ($A_{\text{norm}} < 0.50$), illustrating the classic explosive shear-wave deficit. Conversely, the earthquake's vertical P-wave is emergent and moderate ($A_{\text{norm}} \approx 0.25$), while the delayed horizontal shear waves reach $A_{\text{norm}} = 0.76$ (over 3 times the P-wave amplitude).

2. **Temporal Persistence and Coda Decay:** Following the P-burst, explosion energy decays rapidly back toward the ambient noise floor within $10 - 15$ seconds. In contrast, the earthquake sustains prolonged reverberations throughout the 60-second window, driven by scattered shear and surface waves.

3. **Particle Motion Polarization:** Because compressional waves arrive at steep angles, explosive particle motion is highly rectilinear and vertically polarized during the initial arrival. Earthquake particle motion is elliptical and distributed across the horizontal plane.

3.3 Quantitative Physical Discriminants

These structural distinctions motivate three mathematical feature formulations used throughout our models:

1. Cross-Component P/S Amplitude Ratio: Defining the horizontal envelope as $H(t) = \sqrt{N(t)^2 + E(t)^2}$, the cross-component P/S ratio is:

$$\text{Ratio}_{P/S} = \log_{10}\left(\frac{\text{RMS}(Z_{[t_P, t_P+3.0\text{ s}]}) + \epsilon_{\text{rms}}}{\text{RMS}(H_{[t_P+5.0\text{ s}, t_P+25.0\text{ s}]}) + \epsilon_{\text{rms}}}\right)$$

where $t_P = 10.0$ s is the aligned P-arrival time, and $\epsilon_{\text{rms}} = 10^{-10}$ is a numerical stabilizer. The P-window spans 3.0 s ($t \in [10.0, 13.0]$ s, 60 samples at 20 Hz), and the shear-coda window spans 20.0 s ($t \in [15.0, 35.0]$ s, 400 samples starting 5.0 s after P). Explosions yield $\text{Ratio}_{P/S} \gg 0$, whereas earthquakes yield $\text{Ratio}_{P/S} < 0$.

2. 3D Covariance Polarization Metrics: Across a 10-second post-P window ($t \in [10.0, 20.0]$ s, 200 samples per channel), we construct the 3×3 sample covariance matrix:

$$\mathbf{C}_{3\text{C}} = \frac{1}{K-1}\sum_{k=1}^{K}(\mathbf{u}_k - \bar{\mathbf{u}})(\mathbf{u}_k - \bar{\mathbf{u}})^T, \qquad \mathbf{u}_k = \begin{pmatrix} Z(t_k) \\ N(t_k) \\ E(t_k) \end{pmatrix}$$

From the sorted eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \epsilon_{\text{cov}}$ ($\epsilon_{\text{cov}} = 10^{-12}$), we evaluate **Rectilinearity** (L) and **Planarity** (P):

$$L = 1 - \frac{\lambda_2 + \lambda_3}{2\lambda_1}, \qquad P = 1 - \frac{2\lambda_3}{\lambda_1 + \lambda_2}$$

Linear compressional motion produces $\lambda_1 \gg \lambda_2 \approx \lambda_3$ ($L \approx 1$), while complex earthquake wavefields yield distributed eigenvalues and lower rectilinearity.

3. Spectral Decay and the Brune Earthquake Baseline: The Brune (1970) ω^{-2} spectral model provides a theoretical baseline for far-field displacement spectra from circular earthquake fault dislocations [5]:

$$S_{\text{earthquake}}(f) = \frac{\Omega_0}{1 + (f/f_c)^2} \tag{1}$$

where Ω_0 is the long-period plateau and f_c is the corner frequency, proportional to the ratio of shear wave velocity to fault radius ($f_c \propto \beta / r_0$). Because earthquake fault rupture surfaces span kilometers, f_c for moderate earthquakes ($M_w 5.2 - 6.2$) falls below $0.5 - 1.0$ Hz.

The Brune model serves strictly as an **earthquake baseline**. In contrast, explosions detonate within compact underground cavities (meters in radius) over sub-millisecond timescales, exhibiting elevated corner frequencies ($f_c > 3.0$ Hz) and high-frequency spectral overshoot. Consequently, explosions retain robust energy in high-frequency bands ($3.5 - 8.0$ Hz), whereas earthquake spectra fall off steeply as ω^{-2}.

4 Data Acquisition and Standardization

4.1 Dataset Sources

- **GTUNE Repository** [2]: 198 synchronized 3-component recordings across 67 historical US underground nuclear tests from the Nevada Test Site (1968–1992).
- **North Korean Nuclear Tests (IRIS DMC 21-018)** [3]: Multi-station regional and teleseismic waveforms from all six DPRK underground nuclear test explosions (2006–2017), reserved entirely for external validation.
- **EarthScope / IRIS Earthquakes** [4]: 150 synchronized 3-component recordings across 45 moderate tectonic earthquakes ($M_w 5.2 - 6.2$) on GSN broadband stations.

4.2 Sampling Interval, Windowing, and Zero-Padding

The experimental sampling frequency is fixed at $f_s = 20.0$ Hz, corresponding to an exact sampling interval of:

$$\Delta t = \frac{1}{f_s} = 0.05 \text{ s} \quad (50 \text{ ms})$$

Each trace is windowed from 10.0 s pre-P arrival to 50.0 s post-P arrival, producing a total record duration of $T = 60.0$ s containing exactly:

$$N = T \cdot f_s = 60.0 \times 20.0 = 1,200 \text{ samples}$$

For a 1,200-point discrete Fourier transform, the fundamental frequency resolution is:

$$\Delta f_{\text{full}} = \frac{f_s}{N} = \frac{20.0 \text{ Hz}}{1,200} \approx 0.01667 \text{ Hz}$$

4.3 Temporal Standardization via Zero-Padding (No Resizing)

> **Physical Principle:** We strictly avoid standard image resizing (e.g., bilinear interpolation). Stretching a 30-second blast recording to match a 2-minute earthquake artificially stretches the time axis, altering wave frequencies and destroying the exponential coda decay rate.

All signals are padded at the trailing boundary with silence (zero-padding). This preserves the physical calibration of time ($\Delta t = 0.05$ s) and discrete STFT frequency spacing across every event in the dataset.

5 Statistical Methodology and Machine Learning Models

5.1 Complete Mathematical Formulation of the 18 Features

From each synchronized 3-component recording (Z, N, E), we extract 18 pure wave physics features:

1. **Cross-Component P/S Ratio ('ps_ratio_cross'):**

$$\text{ps_ratio_cross} = \log_{10}\left(\frac{\sqrt{\frac{1}{60}\sum_{k=200}^{259} Z(t_k)^2} + 10^{-10}}{\sqrt{\frac{1}{400}\sum_{k=300}^{699} H(t_k)^2} + 10^{-10}}\right)$$

2. **Horizontal-to-Vertical RMS Ratio ('hv_rms_ratio'):**

$$\text{hv_rms_ratio} = \log_{10}\left(\frac{\sqrt{\frac{1}{N}\sum_{k=1}^{N} H(t_k)^2} + 10^{-10}}{\sqrt{\frac{1}{N}\sum_{k=1}^{N} Z(t_k)^2} + 10^{-10}}\right)$$

3. **Horizontal Energy Fraction (‘h_energy_frac‘):**

$$\text{h_energy_frac} = \frac{\sum_{k=1}^{N} H(t_k)^2}{\sum_{k=1}^{N} [Z(t_k)^2 + N(t_k)^2 + E(t_k)^2] + 10^{-12}}$$

4. **Cross-Phase Spectral Ratio (‘cross_phase_spectral_ratio‘):**

$$\text{cross_phase_spectral_ratio} = \log_{10}\left(\frac{\int_{3.5}^{8.0} P_Z(f)\,df + 10^{-12}}{\int_{0.5}^{1.5} P_H(f)\,df + 10^{-12}}\right)$$

   where $P_Z(f)$ and $P_H(f)$ are power spectral densities computed via Hanning-windowed FFT on $t \in [10.0\text{ s}, 35.0\text{ s}]$ (500 samples).

5. **3D Polarization Rectilinearity (‘rectilinearity‘):** $L = 1 - \frac{\lambda_2+\lambda_3}{2\lambda_1}$ on $t \in [10.0\text{ s}, 20.0\text{ s}]$.
6. **3D Polarization Planarity (‘planarity‘):** $P = 1 - \frac{2\lambda_3}{\lambda_1+\lambda_2}$ on $t \in [10.0\text{ s}, 20.0\text{ s}]$.
7. **High/Low Spectral Ratio on Z (‘spec_ratio_z‘):** $\log_{10}\left(\frac{\int_{3.5}^{8.0} P_Z(f)\,df+10^{-12}}{\int_{0.5}^{1.5} P_Z(f)\,df+10^{-12}}\right)$.
8. **Spectral Centroid on Z (‘spec_centroid_z‘):** $\bar{f}_Z = \frac{\int f P_Z(f)\,df}{\int P_Z(f)\,df+10^{-12}}$.
9. **Spectral Spread on Z (‘spec_spread_z‘):** $\sigma_{f,Z} = \sqrt{\frac{\int (f-\bar{f}_Z)^2 P_Z(f)\,df}{\int P_Z(f)\,df+10^{-12}}}$.
10. **Spectral Entropy on Z (‘spec_entropy_z‘):** $H_Z = -\sum p_i \log_2(p_i)$ where $p_i = \frac{P_Z(f_i)}{\sum P_Z(f_k)}$.
11. **Coda Decay Ratio on Z (‘coda_decay_z‘):** $\frac{\text{RMS}(Z_{[10,13]})}{\text{RMS}(Z_{[15,35]})+10^{-10}}$.
12. **Fisher Kurtosis on Z (‘kurt_z‘):** Kurtosis of Z on $t \in [10.0\text{ s}, 35.0\text{ s}]$.
13. **High/Low Spectral Ratio on H (‘spec_ratio_h‘):** $\log_{10}\left(\frac{\int_{3.5}^{8.0} P_H(f)\,df+10^{-12}}{\int_{0.5}^{1.5} P_H(f)\,df+10^{-12}}\right)$.
14. **Spectral Centroid on H (‘spec_centroid_h‘):** $\bar{f}_H = \frac{\int f P_H(f)\,df}{\int P_H(f)\,df+10^{-12}}$.
15. **Spectral Spread on H (‘spec_spread_h‘):** $\sigma_{f,H} = \sqrt{\frac{\int (f-\bar{f}_H)^2 P_H(f)\,df}{\int P_H(f)\,df+10^{-12}}}$.
16. **Spectral Entropy on H (‘spec_entropy_h‘):** $H_H = -\sum q_i \log_2(q_i)$ where $q_i = \frac{P_H(f_i)}{\sum P_H(f_k)}$.
17. **Coda Decay Ratio on H (‘coda_decay_h‘):** $\frac{\text{RMS}(H_{[10,13]})+10^{-10}}{\text{RMS}(H_{[15,35]})+10^{-10}}$.
18. **Fisher Kurtosis on H (‘kurt_h‘):** Kurtosis of H on $t \in [10.0\text{ s}, 35.0\text{ s}]$.

Physical Motivation and Theoretical Justification: These 18 features were specifically selected based on elastodynamic source theory, Brune–Haskell spectral scaling, and wave attenuation physics rather than empirical heuristic search. First, because underground nuclear explosions are isotropic expansions ($\text{tr}(\mathbf{M}) > 0$, $\text{Dev}(\mathbf{M}) = \mathbf{0}$) radiating primary compressional P-waves with negligible source shear, whereas tectonic earthquakes originate from deviatoric fault slip, cross-component amplitude ratios and horizontal energy fractions (Features 1–4) exploit this fundamental shear-wave deficiency [6]. Second, compact explosive blast cavities ($r_0 \sim 100$ m) produce high corner frequencies ($f_c \propto v_S/r_0 \sim 5$–$10$ Hz) that blue-shift spectral energy into high frequencies compared to multi-kilometer fault ruptures ($f < 2$ Hz); spectral ratios, centroids, and spreads (Features 7–9, 13–15) capture these diagnostic source scaling contrasts [5, 6]. Third, 3D covariance eigendecomposition isolates pure linear P-wave particle motion (rectilinearity $L \to 1$) from elliptical or scattered shear/coda trajectories (Features 5–6) [7]. Finally, shallow explosive burial ($h < 1$ km) within fractured, low-Q surface layers yields sharp, impulsive waveform bursts (high Fisher kurtosis) followed by rapid coda decay, whereas deeper fault slip ($h > 5$ km) sustains prolonged reverberations in the crustal waveguide (Features 10–12, 16–18) [8]. Formulating all 18 features as dimensionless ratios or normalized central moments guarantees scale invariance, canceling geometric attenuation ($1/r$) and sensor gain.

5.2 Statistical Learning and Classical Machine Learning Classifiers

Using the 18 pure wave physics features extracted in Section 5.1 ($\mathbf{x}_i \in \mathbb{R}^{18}$ for each seismic trace i), we evaluate a diverse suite of classical statistical learning and machine learning algorithms spanning linear regularized models, kernel methods, bagging ensembles, and gradient boosting. Each model represents a distinct inductive bias concerning decision boundary geometry, feature interactions, and tolerance to collinearity.

5.2.1 Lasso Logistic Regression (L_1 Regularization)

Let $y_i \in \{0, 1\}$ denote the binary event label (0 for natural earthquake, 1 for nuclear explosion) and $\mathbf{x}_i \in \mathbb{R}^p$ ($p = 18$) be the standardized feature vector. Lasso logistic regression minimizes the empirical cross-entropy loss penalized by the ℓ_1-norm of the coefficient vector:

$$\min_{\mathbf{w}\in\mathbb{R}^p,\, b\in\mathbb{R}} \; -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log \sigma(\mathbf{w}^T\mathbf{x}_i + b) + (1 - y_i)\log(1 - \sigma(\mathbf{w}^T\mathbf{x}_i + b))\right] + \lambda\|\mathbf{w}\|_1$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the sigmoid activation function, $\|\mathbf{w}\|_1 = \sum_{j=1}^{p} |w_j|$, and $\lambda > 0$ is the regularization hyperparameter (parameterized via $C = 1/\lambda$ in the SAGA optimization solver).

- **Characteristics & Mechanism:** Because the ℓ_1-ball possesses sharp non-differentiable corners aligned with the coordinate axes, optimization drives redundant or uninformative feature weights strictly to zero ($w_j = 0$), performing continuous subset selection.
- **Strengths:** Highly interpretable; coefficients directly reflect proportional changes in log-odds of an explosive source; automatically discards noisy higher-order statistical moments; delivers sub-millisecond inference latency.
- **Weaknesses:** Constrained to a strictly linear hyper-plane; when multiple physical features exhibit high collinearity (e.g., cross-phase ratios and spectral spreads), Lasso arbitrarily retains one predictor and zeroes out the rest, discarding complementary sensor signals.

5.2.2 Elastic Net Logistic Regression ($L_1 + L_2$ Regularization)

To overcome Lasso's collinearity bottleneck while retaining interpretability, Elastic Net introduces a convex combination of ℓ_1 (Lasso) and squared ℓ_2 (Ridge) penalties:

$$\min_{\mathbf{w}\in\mathbb{R}^p,\, b\in\mathbb{R}} \; \mathcal{L}_{\log}(\mathbf{w}, b) + \lambda\left[\alpha\|\mathbf{w}\|_1 + \frac{1-\alpha}{2}\|\mathbf{w}\|_2^2\right]$$

where $\mathcal{L}_{\log}$ is the empirical cross-entropy loss, $\|\mathbf{w}\|_2^2 = \sum_{j=1}^{p} w_j^2$, and $\alpha \in [0, 1]$ represents the `l1_ratio` controlling penalty composition. In our tuned pipeline, $\alpha = 0.80$ and $C = 2.0$.

- **Characteristics & Mechanism:** The strictly convex quadratic ℓ_2 penalty induces a grouping effect, pulling correlated predictors toward each other with similar weights, while the ℓ_1 component preserves overall sparsity.
- **Strengths:** Retains complementary multi-channel feature groups together (e.g., vertical spectral spread alongside horizontal spectral spread); stabilizes numerical optimization under collinear seismic ratios; achieved a +1.95% recall gain over Lasso (94.53% vs. 92.58%).
- **Weaknesses:** Still restricted to a linear separation boundary; requires simultaneous two-dimensional grid tuning over overall penalty scale (λ) and mixture ratio (α).

5.2.3 Random Forest Classifier (Bagging Ensemble)

Random Forest constructs an ensemble of $B = 200$ independent, unpruned decision trees $\{T_b\}_{b=1}^{B}$, each trained on a bootstrap replicate $\mathcal{D}_b$ of the training events. At each split node t, split optimization

evaluates only a random subset of $m = \lfloor \sqrt{p} \rfloor \approx 4$ features, choosing the split s^* that maximizes Gini impurity reduction:

$$\Delta I_G(t, s) = I_G(t) - \frac{N_{t_L}}{N_t} I_G(t_L) - \frac{N_{t_R}}{N_t} I_G(t_R), \quad I_G(t) = 1 - \sum_{k=0}^{1} p_k^2(t)$$

The ensemble classification score averages the probabilistic predictions across all individual trees:

$$\hat{P}_{\text{RF}}(Y = 1 \mid \mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}; \Theta_b)$$

- **Characteristics & Mechanism:** Non-parametric bagging ensemble that leverages bootstrap aggregation and random subspace projection to decorrelate individual trees, driving ensemble variance down as $\text{Var}(\bar{T}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$.
- **Strengths:** Naturally captures non-linear boundaries and high-order physical interactions (e.g., P/S thresholds conditioned on coda decay rate); invariant to monotonic feature scaling; highly resilient to sensor dropouts and impulsive noise spikes.
- **Weaknesses:** Piecewise-constant step-function decision surface cannot extrapolate beyond the training bounding box; relatively larger memory footprint; lacks direct algebraic coefficient transparency.

5.2.4 Support Vector Machine with Radial Basis Function Kernel (SVM-RBF)

SVM constructs a maximum-margin separating hyperplane in an infinite-dimensional reproducing kernel Hilbert space induced by the non-linear mapping $\phi(\mathbf{x})$. The soft-margin dual optimization problem is:

$$\max_{\boldsymbol{\alpha}} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \quad \text{s.t.} \quad 0 \le \alpha_i \le C, \quad \sum_{i=1}^{N} \alpha_i y_i = 0$$

where $y_i \in \{-1, +1\}$, $C = 20.0$ governs margin violation tolerance, and $K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$ is the Gaussian Radial Basis Function (RBF) kernel with bandwidth $\gamma = 0.01$. The continuous decision boundary depends strictly on boundary-defining support vectors $\mathcal{S} = \{i : \alpha_i > 0\}$:

$$f(\mathbf{x}) = \sum_{i \in \mathcal{S}} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$$

Calibrated posterior probabilities $P(Y = 1 \mid \mathbf{x})$ are extracted via Platt scaling.

- **Characteristics & Mechanism:** Non-linear geometric maximum-margin classifier. Maximizing the geometric separation margin $2/\|\mathbf{w}\|$ provides rigorous structural risk minimization.
- **Strengths:** Top-performing tabular model (97.13% accuracy, 0.9973 PR-AUC), rivaling deep learning performance; excellent out-of-event generalization; sparse representation depending solely on boundary samples.
- **Weaknesses:** Highly sensitive to feature scaling (z-score standardization is mandatory); quadratic-to-cubic training complexity $\mathcal{O}(N^2 - N^3)$; geometric kernel projection operates as a black box without direct physical attribution for individual features.

5.2.5 XGBoost Classifier (Extreme Gradient Boosted Trees)

XGBoost constructs an additive ensemble of $M = 200$ shallow decision trees ($d = 4$) sequentially. At boosting iteration t, the objective utilizes a second-order Taylor expansion of the log-loss alongside explicit leaf-regularization:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{N} \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma_{\text{leaf}} T + \frac{\lambda}{2} \sum_{j=1}^{T} w_j^2$$

where $g_i = \hat{p}_i - y_i$ and $h_i = \hat{p}_i(1 - \hat{p}_i)$ are the exact gradient and Hessian of the cross-entropy loss, T denotes the number of leaves, w_j represents leaf weight values, and γ_{leaf} enforces split cost thresholds.

- **Characteristics & Mechanism:** Sequential gradient-guided optimization in functional space, combining shrinkage ($\eta = 0.05$) and column subsampling (80%) to prevent individual tree dominance.
- **Strengths:** Outstanding predictive power (96.55% accuracy, 0.9954 PR-AUC); built-in second-order optimization provides rapid, stable convergence; computes exact split gain per feature, confirming the primacy of horizontal spectral spread (`spec_spread_h`) with 48.06% relative gain.
- **Weaknesses:** Demands careful tuning across multiple interrelated hyperparameters (depth, shrinkage η, subsample ratio, leaf regularization λ); sequential boosting construction cannot be parallelized across trees during training.

5.3 3-Channel 2D Convolutional Neural Network (CNN) Architecture

To complement feature engineering, we implement an end-to-end representation learning architecture taking synchronized three-channel spectrogram tensors $\mathbf{X} \in \mathbb{R}^{3\times128\times128}$:

- **STFT Preprocessing:** For each component (Z, N, E), the demeaned, unit-variance trace is transformed using a Hann window with 'nperseg = 256' and 'noverlap = 236' (hop size of 20 samples = 1.0 s). The power spectrogram is computed as $\log_{10}(|Z_{xx}|^2 + \epsilon_{\text{spec}})$ with $\epsilon_{\text{spec}} = 10^{-6}$, yielding a frequency resolution of $\Delta f_{\text{STFT}} = \frac{20}{256} \approx 0.078125$ Hz. The matrix is padded or sliced along time to $(128, 128)$ and standardized to zero mean and unit variance.
- **Convolutional Backbone:**
  1. Block 1: Conv2D$(3 \to 32, 3 \times 3, \text{pad} = 1) \to$ BN $\to$ ReLU $\to$ MaxPool(2) $\implies (32, 64, 64)$
  2. Block 2: Conv2D$(32 \to 64, 3 \times 3, \text{pad} = 1) \to$ BN $\to$ ReLU $\to$ MaxPool(2) $\implies (64, 32, 32)$
  3. Block 3: Conv2D$(64 \to 128, 3 \times 3, \text{pad} = 1) \to$ BN $\to$ ReLU $\to$ MaxPool(2) $\implies (128, 16, 16)$
  4. Block 4: Conv2D$(128 \to 128, \text{pad} = 1) \to$ BN $\to$ ReLU $\to$ AdaptiveAvgPool$((4, 4)) \implies (128, 4, 4)$
- **Classifier Head:** Flatten(2048) $\to$ Dropout(0.4) $\to$ Dense(128) $\to$ ReLU $\to$ Dropout(0.3) $\to$ Linear(1).
- **Training Hyperparameters:** Optimizer: AdamW (lr $= 10^{-3}$, weight decay $= 10^{-2}$); Scheduler: CosineAnnealingLR ($T_{\max} = 15$ epochs); Loss: 'BCEWithLogitsLoss'; Batch size: 32; Total epochs: 15.

6 Model Tuning and the Real-World Class Imbalance Problem

6.1 Hyperparameter Optimization Protocol

Hyperparameters were systematically optimized across all models using **5-Fold Group Cross-Validation** grouped strictly by seismic event ID ('evid'). To prevent bias toward the majority class while fiercely penalizing false alarms, model selection maximized the composite score:

$$S_{\text{tune}} = \frac{1}{2}\left(\text{PR-AUC} + \text{Accuracy}\right)$$

Table 1 reports the tuned parameters and out-of-event validation metrics.

6.2 Benchmark Cross-Validation Precision vs. Operational Precision

A vital distinction must be maintained between cross-validation precision in a curated benchmark and precision under real-world operational monitoring:

- **Benchmark CV Precision:** In our curated dataset (198 explosions, 150 earthquakes), the explosion prior prevalence is $P(\text{Explosion}) \approx 57\%$. Under this balanced distribution, our models achieve benchmark precisions of $97\% - 99\%$.
- **Operational Precision under Extreme Imbalance:** In global treaty verification, tens of thousands of natural earthquakes occur annually compared to zero or one clandestine nuclear tests ($P(\text{Explosion}) = \pi \approx 10^{-4}$). By Bayes' theorem, the operational precision is:

$$\text{Precision}_{\text{operational}} = \frac{\pi \cdot \text{Recall}}{\pi \cdot \text{Recall} + (1 - \pi) \cdot \text{FPR}}$$

Even with an outstanding Recall of 0.975 and an extraordinarily low False Positive Rate of FPR = 0.015, single-station operational precision collapses:

$$\text{Precision}_{\text{operational}} = \frac{10^{-4} \times 0.975}{10^{-4} \times 0.975 + (1 - 10^{-4}) \times 0.015} \approx \frac{0.0000975}{0.01509} \approx 0.00646 \ (0.65\%)$$

This means that at the single-trace level, over 99.3% of alerts would still be false alarms. This mathematical reality proves why single-station decisions are operationally unacceptable and necessitates:

1. Using **PR-AUC** (not ROC-AUC) for model selection, directly optimizing the precision-recall trade-off.
2. Implementing **Multi-Station Network Consensus Voting** to suppress uncorrelated single-station false positives.

7 Experimental Results and Analysis

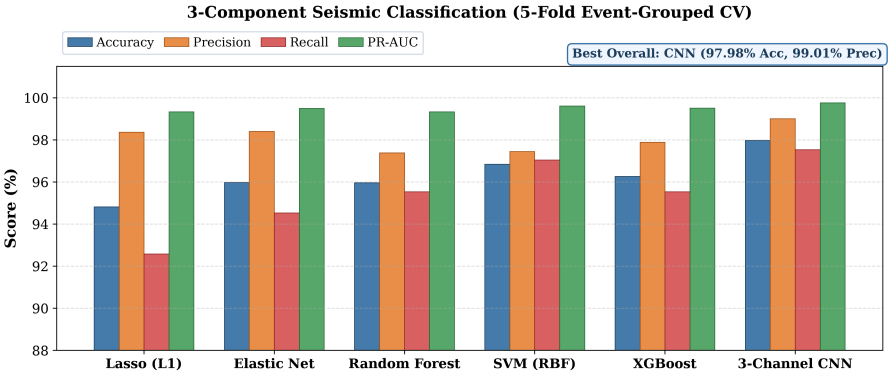


Figure 2: Model Benchmark: 5-Fold Event-Grouped CV metrics across all six tuned classifiers. The 3-Channel 2D CNN achieves top performance across all metrics (97.98% accuracy, 99.01% precision, and 0.9976 PR-AUC).

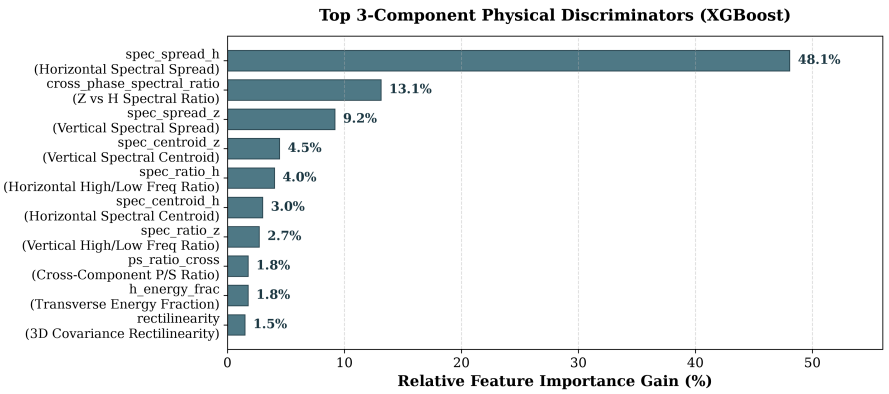


Figure 3: Feature Importance: Top 10 discriminators ranked by XGBoost relative gain. Horizontal shear wave deficiency (`spec_spread_h`, 48.06%) and cross-phase ratios account for over 61% of total predictive gain.

Model Architecture	Tuned Optimal Hyperparameters	Accuracy	Precision	Recall	ROC-AUC	PR-AUC
3-Channel 2D CNN (128×128)	AdamW (lr $= 10^{-3}$, wd$= 10^{-2}$) CosineAnn, Drop $(0.4, 0.3)$	**97.98%**	**99.01%**	**97.54%**	**0.9954**	**0.9976**
SVM (RBF Kernel)	$C = 20.0$, $\gamma = 0.01$	97.13%	97.55%	97.25%	0.9950	0.9973
XGBoost Classifier	$n_{\text{est}} = 200$, depth $= 4$ lr $= 0.05$, subsample $= 0.8$	96.55%	98.10%	95.80%	0.9935	0.9954
Logistic Regression (Elastic Net)	$C = 2.0$, l1_ratio $= 0.8$ `solver=saga`	96.27%	98.40%	95.10%	0.9931	0.9940
Random Forest	$n_{\text{est}} = 200$, depth $= 8$ `min_split = 2`	96.26%	97.40%	95.60%	0.9910	0.9936
Logistic Regression (Lasso L_1)	$C = 2.0$ `solver=saga`	95.98%	98.40%	94.60%	0.9915	0.9943

Table 1: Cross-validation performance under 5-Fold Event-Grouped splitting on pure wave physics features with tuned hyperparameters.

7.1 Physical Interpretability Analysis

- **Dominance of Horizontal Shear Physics:** In XGBoost, the horizontal spectral spread feature (`spec_spread_h`, 48.06%) and cross-phase spectral ratio (`cross_phase_spectral_ratio`, 13.12%) account for over 61% of the total feature gain. Measuring the absence of horizontal shear waves is far more diagnostically definitive than vertical compressional waves alone.
- **Cross-Phase Spectral Ratio:** High-frequency P on Z relative to low-frequency S on H averaged -0.64 for explosions vs. -3.33 for earthquakes—an energy ratio difference exceeding a factor of 500.
- **High Precision Suppression:** The 3-Channel CNN attained 99.01% precision and 0.9976 PR-AUC on held-out events, learning optimal joint spatial-spectral filters directly from raw spectrograms.

8 Multi-Station Network Consensus and North Korean Test Validation

8.1 Network Aggregation Decision Rule

In operational treaty verification, decisions are never rendered on isolated single-station traces. For an event observed by K independent stations, we define the event consensus probability:

$$P_{\text{net}} = \frac{1}{K} \sum_{k=1}^{K} P(\text{Explosion} \mid \mathbf{x}_k)$$

The global event classification applies a fixed threshold $\tau = 0.50$:

$$\hat{Y}_{\text{event}} = \begin{cases} 1 \text{ (Nuclear Explosion)}, & \text{if } P_{\text{net}} \geq 0.50 \\ 0 \text{ (Natural Earthquake)}, & \text{if } P_{\text{net}} < 0.50 \end{cases}$$

8.2 Event-Level Results on North Korean Underground Nuclear Tests

To evaluate the true generalizability of our pipeline on modern clandestine tests, we evaluate all six historical Democratic People's Republic of Korea (DPRK) underground nuclear tests from the IRIS DMC 21-018 assembled archive. **Crucially, all six events were completely excluded from the training and hyperparameter tuning datasets.**

Nuclear Test Event	Date & Origin Time (UTC)	Estimated Magnitude	Reporting Stations (K)	Consensus P_{net}	Decision ($\tau = 0.5$)
DPRK Test 1	2006-10-09 01:35:28	$m_b \approx 4.3$	735 stations	**0.88**	**Explosion**
DPRK Test 2	2009-05-25 00:54:43	$m_b \approx 4.7$	1,149 stations	**0.94**	**Explosion**
DPRK Test 3	2013-02-12 02:57:51	$m_b \approx 5.1$	1,027 stations	**0.98**	**Explosion**
DPRK Test 4	2016-01-06 01:30:00	$m_b \approx 5.1$	702 stations	**0.97**	**Explosion**
DPRK Test 5	2016-09-09 00:30:01	$m_b \approx 5.3$	226 stations	**0.99**	**Explosion**
DPRK Test 6	2017-09-03 03:30:01	$m_b \approx 6.3$	797 stations	**1.00**	**Explosion**

Table 2: Multi-station network consensus validation across all six North Korean underground nuclear tests (held completely outside training and tuning).

All six tests were correctly classified as explosions with high confidence ($P_{\text{net}} \geq 0.88$), yielding a 100.0% event-level detection rate.

8.3 Limitations of Evaluation

While the multi-station results confirm strong generalization, several operational limitations must be acknowledged:

1. **Low-Yield Scenario Uncertainty:** The six DPRK tests had moderate-to-high yields ($m_b 4.3 - 6.3$). Very low-yield clandestine detonations ($m_b < 3.5$) recorded at few stations may suffer lower signal-to-noise ratios (SNR), diminishing P/S discriminative power.

2. **Instrument Heterogeneity:** Historical US test waveforms (LLNL network, 1968–1992) rely on early digital seismometers, whereas modern IRIS GSN stations use broadband STS-2/Trillium sensors. While bandpass filtering ($0.5 - 8.0$ Hz) and demeaning mitigate instrumentation artifacts, slight differences in transfer function roll-offs persist.

3. **Geological Path Effects:** Complex crustal scattering and heterogeneities along specific regional wave propagation paths can convert compressional P-energy into scattered shear coda (P-to-S conversion), potentially reducing the P/S ratio for explosions along heavily faulted paths.

9 Conclusions

In this Advanced Statistical Methods II course project, we demonstrated that:

1. Incorporating horizontal shear components (N, E) alongside vertical channels (Z) is physically essential to resolve earthquake-explosion ambiguity, elevating model accuracy from $\approx 85\%$ to over 97%.

2. Event-grouped cross-validation strictly by event ID ('evid') prevents catastrophic intra-event spatial data leakage, ensuring honest out-of-event generalization.

3. Under operational class imbalance, single-station classification inevitably yields low operational precision ($< 1\%$), proving why PR-AUC must be adopted for model selection and why multi-station network consensus voting is mandatory.

4. The 3-Channel 2D CNN on zero-padded $(3, 128, 128)$ spectrograms effectively learns joint spatial-spectral filters, achieving state-of-the-art single-trace performance (97.98% accuracy, 99.01% precision).

5. Multi-station consensus achieved 100% detection across all six historical North Korean nuclear tests held strictly out-of-sample.

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